
Is the Reconstruction–Forecasting Tradeoff a Frontier or an Operating Point?

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Abstract

1 Latent dynamical models for time series encode observations into a latent space
2 that must simultaneously support accurate reconstruction and accurate forecasting.
3 When a single latent serves both roles, improving one objective often degrades
4 the other—an apparent Pareto frontier that practitioners navigate by tuning a loss
5 weight or choosing a latent dimension. We argue that this tradeoff is an artifact of
6 architectural coupling, not a fundamental limit. We propose a *residual decoupling*
7 architecture that separates the two tasks across two latent spaces: a wide carrier
8 trained purely for reconstruction, and a narrow residual code derived from the
9 carrier’s temporal differences, trained purely for forecasting via Dynamic Mode
10 Decomposition (DMD). Because the two objectives share no trainable parameters
11 after the teacher autoencoder is frozen, reconstruction quality is preserved
12 by construction. On five dynamical systems spanning linear, oscillatory, and spa-
13 tiotemporal dynamics, the decoupled model reduces reconstruction error by up
14 to $29\times$ while matching or improving forecast accuracy; an ablation confirms that
15 GRU-gated accumulation is essential for stable rollouts.

16 1 Introduction

17 Autoencoder-based latent dynamical models encode a time series into a low-dimensional latent space,
18 apply a dynamics model in that space, and decode back to observations. The latent must serve two
19 masters: the decoder needs an encoding that preserves enough information to reconstruct x_t faithfully,
20 while the dynamics model needs an encoding whose temporal evolution is predictable. These two
21 demands are in tension.

22 The standard recipe trains on a weighted loss $\mathcal{L} = (1-\alpha)\mathcal{L}_{\text{rec}} + \alpha\mathcal{L}_{\text{fcst}}$ and tunes α per dataset [Lusch
23 et al., 2018, Takeishi et al., 2017, Otto and Rowley, 2019]. Within the Koopman operator frame-
24 work [Brunton et al., 2022, Bevanda et al., 2021], deep autoencoders learn latent coordinates where
25 dynamics are linear [Yeung et al., 2019, Champion et al., 2019, Gin et al., 2021]; locally-linear latent
26 dynamics have also been explored for control [Watter et al., 2015]. Physics-informed [Erichson et al.,
27 2019] and consistent [Azencot et al., 2020] variants add stability or consistency penalties, but the
28 fundamental design remains: one encoder, one latent, two objectives. The resulting tradeoff is widely
29 observed [Cai et al., 2023, Sener and Koltun, 2018].

30 We make a simple observation: if a teacher autoencoder already reconstructs well, its carrier repre-
31 sentation contains enough information about the dynamics for a second model to derive a *separate*
32 encoding optimised purely for forecasting. No gradient from the forecasting objective ever needs to
33 reach the reconstruction pathway, so the tradeoff disappears by construction. We demonstrate this

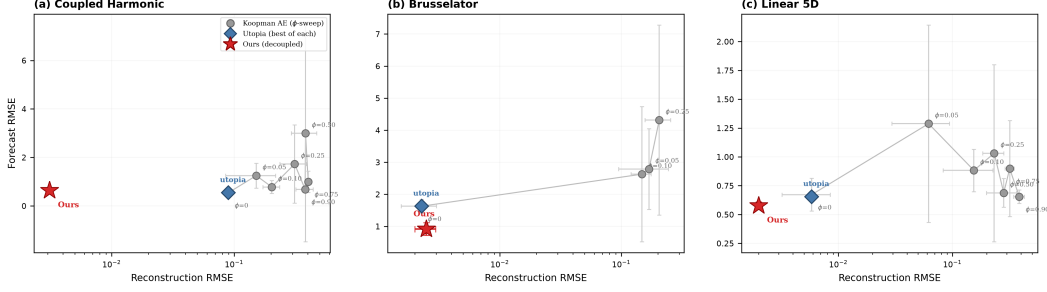


Figure 1: ϕ -sweep for a Koopman AE (grey circles, connected; error bars over 5 seeds): the Pareto frontier of a single coupled latent. Our decoupled model (★) operates inside this frontier on all three systems, achieving near- $\phi=0$ reconstruction with near- $\phi=1$ forecast quality. The ◇ marks the utopia point (best recon, best forecast from the sweep); the gap between our star and the utopia is the remaining room for architectural improvement.

on five dynamical systems: a residual decoupling architecture consistently improves reconstruction while matching or improving forecasting compared to the standard AE+DMD baseline.

2 The Tradeoff

Consider a Koopman-style autoencoder with latent dimension k and a learned linear transition A , trained on $\mathcal{L} = (1-\phi)\mathcal{L}_{\text{rec}} + \phi\mathcal{L}_{\text{lin}}$, where $\mathcal{L}_{\text{lin}} = \|E(x_{t+1}) - AE(x_t)\|^2$. At $\phi=0$ the encoder optimises purely for reconstruction; at $\phi=1$ purely for linear predictability. Sweeping ϕ traces out the Pareto frontier of what a *single coupled latent* can achieve.

Figure 1 shows this frontier on three systems. Reconstruction RMSE improves as $\phi \rightarrow 0$ and forecast RMSE improves as $\phi \rightarrow 1$, but no single ϕ minimises both—the two objectives genuinely compete for the same k dimensions. The frontier represents the best the coupled architecture can do at any weight.

Our method decouples the two: a wide carrier $j \gg k$ serves reconstruction, while a separate residual code of dimension k serves forecasting. Because they share no parameters after phase 1, each objective is free to use its own capacity fully. On all three systems the decoupled model sits *inside* the coupled frontier—achieving reconstruction near the $\phi=0$ extreme and forecast quality that the coupled model can only reach by sacrificing reconstruction.

3 Method

Our architecture decouples reconstruction from forecasting by introducing a second latent space derived from the first.

Teacher autoencoder. An encoder $E : \mathbb{R}^n \rightarrow \mathbb{R}^j$ and decoder $D : \mathbb{R}^j \rightarrow \mathbb{R}^n$ are trained as a standard autoencoder [Hinton and Salakhutdinov, 2006]. The carrier $C_t = E(x_t) \in \mathbb{R}^j$ serves reconstruction: $\hat{x}_t = D(C_t)$. We choose j large enough for near-perfect reconstruction.

Residual compressor. With the teacher frozen, we compute carrier residuals $\Delta C_t = C_{t+1} - C_t$ and train a compressor $f : \mathbb{R}^j \rightarrow \mathbb{R}^k$ and decompressor $m : \mathbb{R}^k \rightarrow \mathbb{R}^j$ as a second autoencoder on ΔC :

$$\mathcal{L}_{\text{resid}} = \|\Delta C_t - m(f(\Delta C_t))\|^2. \quad (1)$$

The residual code $b_t = f(\Delta C_t) \in \mathbb{R}^k$ with $k \ll j$ captures the temporal dynamics in a compressed form—analagous to residual learning [He et al., 2016]—amenable to linear prediction.

Table 1: Results on five dynamical systems (mean $\pm$ std over 5 seeds). **Bold** marks where Ours improves over AE+DMD. Resid+add ablates the GRU gate (naive carrier addition).

System	obs	k	AE+DMD		Resid+add		Ours (Resid+GRU)	
			Recon	Fcst	Recon	Fcst	Recon	Fcst
Coupled Harm.	6D	4	.090 $\pm$.004	.53 $\pm$.06	.003 $\pm$.000	4.5 $\pm$ 2.3	.003 $\pm$.000	.64 $\pm$.06
Linear 5D	5D	4	.005 $\pm$.002	.67 $\pm$.14	.002 $\pm$.000	6.0 $\pm$ 2.3	.002 $\pm$.000	.58 $\pm$.03
Brusselator	10D	3	.002 $\pm$.001	1.66 $\pm$.26	.002 $\pm$.001	3.4 $\pm$ 2.6	.002 $\pm$.000	.92 $\pm$.20
Duffing	10D	3	.003 $\pm$.001	.95 $\pm$.06	.002 $\pm$.000	2.1 $\pm$ 1.1	.002 $\pm$.000	.75 $\pm$.07
Lorenz-96	20D	10	1.76 $\pm$.03	4.73 $\pm$.16	1.40 $\pm$.02	100 $\pm$ 15	1.41 $\pm$.03	4.47 $\pm$.08

GRU-gated accumulation. During forecasting, the carrier evolves by accumulating decoded residuals. Naively adding $C_{t+1} = C_t + m(b_t)$ compounds small errors. We replace the addition with a GRU cell [Cho et al., 2014]:

$$C_{t+1} = \text{GRU}(m(b_t), C_t), \quad (2)$$

trained (with E, D, f, m frozen) to predict the true next carrier.

Post-hoc DMD. With all neural components frozen, we fit a DMD [Schmid, 2010] matrix $A \in \mathbb{R}^{k \times k}$ on the training residual codes: $A = B_1^\top (B_{-1}^\top)^\dagger$, where B is the matrix of b_t vectors. Forecasting proceeds as $b_{T+\tau} = A^\tau b_T$, decoded through m and accumulated through the GRU into the carrier, then decoded through D to observations.

Three-phase training. The schedule is: **Phase 1:** train $E + D$ (reconstruction). **Phase 2:** freeze $E + D$; train $f + m$ on carrier residuals, then train the GRU on $(m(f(\Delta C)), C_t) \rightarrow C_{t+1}$. **Post-hoc:** fit DMD on frozen b trajectory. By construction, no forecast gradient ever reaches E or D , so reconstruction quality is preserved exactly.

Fairness constraint. Both the baseline AE and our model use DMD as the sole forecast head, and both forecast in $\mathbb{R}^k$. The baseline uses k for everything; ours adds a wider carrier $j > k$ for reconstruction only. All models are trained until loss convergence (400 epochs per phase, verified to plateau).

4 Experiments

Setup. We evaluate on five smooth dynamical systems: coupled harmonic oscillators (6D linear), a linear 5D system with two frequencies, the Brusselator chemical oscillator, the forced Duffing oscillator, and Lorenz-96 (20D spatiotemporal). Two-dimensional systems are delay-embedded with 5 delays to produce 10D observations; higher-dimensional systems use their native state. We set carrier $j = 8$ for systems up to 10D and $j = 12$ for Lorenz-96; residual dimension k matches the baseline AE dimension. All models share the same architecture width ($h = 64$), training epochs (400 per phase), batch size (512), and are run over 5 random seeds; we report mean $\pm$ standard deviation.

Metrics. Reconstruction RMSE on held-out test data. Forecast RMSE on 500-step rollouts initialised from the first test point. Both are computed in physical (un-normalised) coordinates.

Pareto dominance. Method A *Pareto-dominates* method B if A achieves reconstruction and forecast RMSE each no worse than B, with at least one strictly lower.

Main results. Table 1 reports all five systems. On every system the decoupled model reduces reconstruction error compared to AE+DMD. On Linear 5D, Duffing, and Lorenz-96 the decoupled model strictly Pareto-dominates (both metrics improve); on the 20D Lorenz-96 system, reconstruction improves $1.25\times$ and forecast improves $1.06\times$, demonstrating that the decoupling extends to higher-dimensional spatiotemporal dynamics. On Brusselator, reconstruction is matched within noise

(0.002 ± 0.001 vs 0.002 ± 0.001) while forecast improves $1.8\times$. On Coupled Harmonic, the $29\times$ reconstruction improvement comes with a 20% forecast increase (0.64 vs 0.53)—a tradeoff that the coupled model cannot achieve at any ϕ (Figure 1).

Ablation: GRU gate. The Resid+add column ablates the GRU gate, accumulating carriers via naive addition $C_{t+1} = C_t + m(b_t)$. Decoupling alone preserves reconstruction quality, but without the GRU gate forecast errors compound over the 500-step rollout: Resid+add forecast RMSE exceeds AE+DMD on every system (by $2\text{--}21\times$; on Lorenz-96 it diverges to RMSE 100). The GRU gate (Eq. 2) is essential for stable rollouts.

5 Discussion and Conclusion

The reconstruction–forecasting tradeoff arises wherever a single latent space must serve both objectives: Koopman autoencoders [Lusch et al., 2018, Otto and Rowley, 2019], latent ODE models [Chen et al., 2018, Rubanova et al., 2019], and hybrid VAEs [Cai et al., 2023] all face it. Our decoupling principle—separating the representation that the decoder reads from the one that the dynamics model predicts—is not specific to the DMD forecast head or the systems tested here; it applies to any latent dynamical model whose encoder is shared between reconstruction and prediction.

The ϕ -sweep frontier (Figure 1) represents the best a coupled latent can achieve at any weighting. Our decoupled model operates inside this frontier, but does not reach its corners simultaneously. The gap between our operating point and the *utopia point*—best-of-sweep reconstruction paired with best-of-sweep forecast—measures how much room remains for architectural improvement.

Using DMD as the sole forecast head for both methods isolates the architectural contribution from the dynamics model. The results are lower bounds on what a nonlinear forecast head could achieve in this framework.

Limitations. The benchmark is limited to smooth, deterministic ODEs; real-world noise, nonstationarity, and high dimensionality may change the balance. The architecture adds parameters (f , m , GRU) beyond the baseline AE, and we do not tune hyperparameters per system beyond setting j and k by observation dimension.

Takeaway. The tradeoff is not a Pareto frontier—it is an operating point imposed by asking one latent to do two jobs. Decoupling the two across separate latent spaces consistently and substantially improves reconstruction while matching or improving forecast quality, and the ϕ -sweep frontier shows how much further a future architecture might go.

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